PV102 Day 1

Applications and Design Decisions

PV102 begins where PV101 ended

PV101 built the system. PV102 decides which system belongs in the job.

Today's rule

Start with the load, the site, and the operating need.

A correct part can still make the wrong system

Architecture must fit the job, not just the available equipment.

Four common PV architectures

Grid-tied. Hybrid. Off-grid. PV-direct.

Grid-tied

PV serves loads through an interconnected utility system.

Hybrid

PV, storage, and the grid work together for selected goals.

Off-grid

The system must carry its loads without utility support.

PV-direct

The load runs when solar power is available; storage may be unnecessary.

Residential and commercial applications

Bill reduction, resilience, available space, rules, and maintenance shape the design.

Remote and critical loads

Reliability, access, weather, storage, and serviceability become dominant constraints.

Constraints are design inputs

Load. Solar resource. Shade. Space. Budget. Access. Maintenance. Rules.

Start with the load

What must run? How many watts? For how long? What starts hard?

The site changes the answer

Sun, shade, orientation, weather, space, and access change system performance.

Storage solves a job and creates new jobs

Batteries add capability, cost, weight, controls, losses, and maintenance.

Protection belongs in the design

Isolation, overcurrent protection, labeling, and verified boundaries are not extras.

Today's challenge

Choose the architecture, map the energy path, and defend the decision.

Team roles

Designer. Verifier. Recorder and handoff lead. Rotate after each scenario.

Acceptance criteria

Match the need. Show the path. Name the constraint. State the safety boundary.

Scenario 1

A remote trail counter needs year-round power where no utility exists.

Scenario 2

A shop wants bill reduction and limited backup for essential loads.

Scenario 3

A water pump only needs to run when strong sunlight is available.

Debrief question 1

Which operating need changed your architecture choice?

Debrief question 2

Which constraint mattered most, and what evidence supports that claim?

Close

The best PV system is the one matched to its real job.